alqa Applied University
lty of Engineering Technology
artment of Electrical Engineering

Electronics I
Final Exam 24-1-2024
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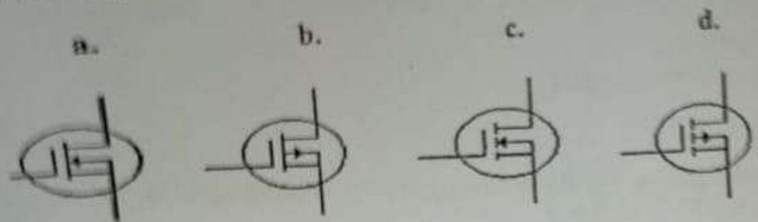
ident Name: الجسم محمود يونس حامد ابومنديل

Instructor Name:

1. Select the correct answer for the following questions:

. The value of α when $\beta = 100$ is .....
- 1.01 b-101 c- 0.99 d- 9.9

2. Which figure represents a p-channel E-MOSFET?



3. In BJT small signal model, the h_{21} parameter unit in hybrid equivalent model is.............
a. Ohms. b. Ampere. c. Volt. d. Unitless.

4. In BJT small signal model, the h_{11} parameter in hybrid equivalent model is calculated by ....
a. Making an open circuit at the input terminals.
b. Making an open circuit at the output terminals.
c. Making a short circuit at the output terminals.
d. Making a short circuit at the input terminals.

5. Using the data sheet for a BJT transistor, the parameter h_{FE} represents ....
a. α_{ac} b. β_{dc} c. β_{ac} d. α_{dc} e. None of them.

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5. Using the data sheet for a BJT transistor, the parameter h_{FE} represents ....

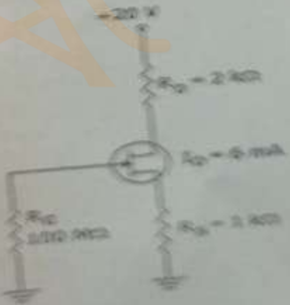
a. α_{ac} b. β_{dc} c. β_{ac} d. α_{dc} e. None of them.

6. The best BJT configuration that provides the best Q- point stability with single-polarity supply voltage is..

a. Emitter follower. b-Collector-feedback. c-Voltage-divider. d- Fixed-bias.

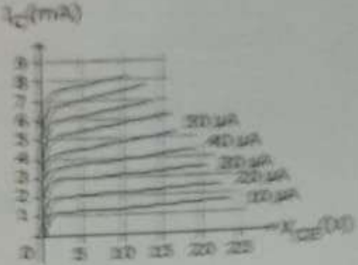
7. Using the figure shown below, the value of Vs is equal to...............

a. 6 V. b. 8 V. c. 20 V. d. 2 V e. None of them.

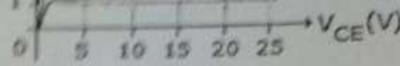


8. In JFET semi-conductor, when V_{DS} is greater than V_P, the JFET has a characteristic of ....
a. Voltage source.	b. Current source.	c. Ohmic region.	d. None of them.

9. For the circuit shown below of collector characteristics, the value of β_{ac} for $V_{CE} = 15\,V$ and $I_B = 30\,\mu A$ is equal to ....
a. 160	b. 100	c. 50.	d. 400.



10. Which of the following is not correct about FET compared to BJT?
a. FET is smaller size than BJT.
b. FET has a higher input impedance.
c. FET has Less temperature stability than BJT.
d. None of the above.

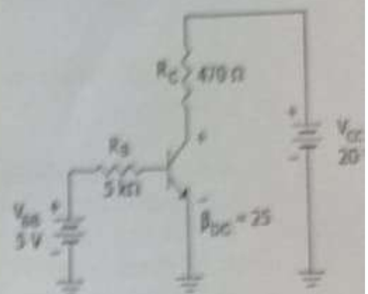


10. Which of the following <u>is not correct</u> about FET compared to BJT?

a. FET is smaller size than BJT.
b. FET has a higher input impedance.
c. FET has Less temperature stability than BJT.
d. None of the above.

11. The collector-emitter voltage V_{CE} for the following configuration is equal to............

a. 19.3 V b-9.2 V c. 0.7 V d- 9.9 V



12. In Shockley's equation for FET, the controlled variable that relates I_D and V_{GS} is......

a. I_{DSS}. b. V_P. c. I_D. d. V_{GS}.

13. For a FET transistor, the I_{DSS} for $V_{GS} = -4.5$ V, $V_p = -8$ V, and $I_D = 6.12$ mA is .....

a. 32 mA b. 16 mA c. 64 mA d. 8 mA

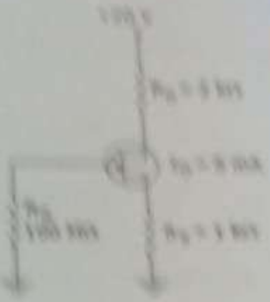
14. The data sheet for E-MOSFET is given below. For VGS (Th) = 3 V, the parameter k of MOSFET is......

ON CHARACTERISTICS

Gate Threshold Voltage ($V_{DS} = 10$ V, $I_D = 10\ \mu A$)	$V_{GS(Th)}$	1.0	5	Vdc
Drain-Source On-Voltage ($I_D = 2.0$ mA, $V_{GS} = 10$V)	$V_{DS(on)}$	–	1.0	V
On-State Drain Current ($V_{GS} = 10$ V, $V_{DS} = 10$ V)	$I_{D(on)}$	3.0	–	mA

a. 6.17 mA/ V^2 b. 6.17 V^2/ mA c. $0.061 * 10^{-3}$ A/V^2 d. $0.061 * 10^{-3}$ V^2/A

15. Using the following figure, the value of V_{CE} is ......
a. 0 V   b. 8 V   c. 2 V   d. 20 V



16. For BJT transistor, the base-emitter junction in saturation region is ......
a. Forward biased   b. Reverse biased   c. Ohmic region   d. None of them

17. Assume that an electronic technician looks for design a circuit that makes the Q- point of the load line of the following circuit is going up toward the I_C, then the value of ...... should be ......
a. R_E, increased   b. R_1, increased   c. R_E, decreased   d. R_1, decreased   e. R_2, increased

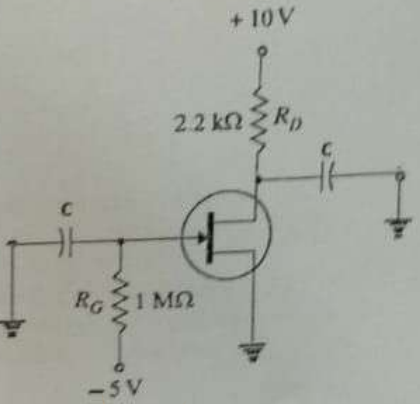
18. The following BJT configuration is called .... configuration.
a. Common base   b. Emitter follower   c. Emitter bias   d. None of them

19. In a pnp transistor, the current carriers are......
a. Acceptor ions   b. donor ions   c. Electrons   d. Holes

20. For the following BJT configuration, the base current is....
a. $I_B = \frac{V_{TH}-V_{BE}}{R_{TH}+(\beta+1)R_E}$ b. $I_B = \frac{V_{TH}-V_E}{R_1+(\beta+1)R_E}$ c. $I_B = \frac{V_{TH}-V_E}{R_{TH}+(\beta+1)R_E}$ d. $I_B = \frac{I_E}{(\beta+1)}$

Q2. For the given circuit below, if $V_{GS(off)} = -8$ V, $I_{DSS} = 16$ mA, and C= 2 μF, answer the following:

1. What is the type of this transistor?



2. What is the name of this configuration?

3. What is the value of V_{GS}?

4. What is the value of I_D.

5. What is the value of V_{DS}?

Multiple Choice Answers of Question 1:

Question	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	C									

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20